

EXPERIMENT 23

Prolog Program: Family Tree

Aim

To write a Prolog program to implement a family tree and determine relationships such as parent, child, sibling, grandparent, uncle, and aunt.

Algorithm

1. Define facts for the male and female members.
2. Define parent facts to represent parent-child relationships.
3. Define rules for child, sibling, grandparent, uncle, and aunt relationships.
4. Give appropriate queries in the Prolog console.
5. Prolog checks the facts and rules and produces the required relationship.

Prolog Code

```
% Family Tree
```

```
% Gender facts
```

```
male(john).
```

```
male(peter).
```

```
male(david).
```

```
male(robert).
```

```
male(james).
```

```
female(mary).
```

```
female(susan).
```

```
female(lisa).
```

```
female(anna).
```

% Parent facts

parent(john, peter).

parent(mary, peter).

parent(john, susan).

parent(mary, susan).

parent(peter, david).

parent(lisa, david).

parent(peter, anna).

parent(lisa, anna).

parent(robert, james).

parent(susan, james).

% Child rule

child(Child, Parent) :-

parent(Parent, Child).

% Sibling rule

sibling(X, Y) :-

parent(P, X),

parent(P, Y),

X \= Y.

% Grandparent rule

grandparent(Grandparent, Grandchild) :-

parent(Grandparent, Parent),

parent(Parent, Grandchild).

% Grandchild rule

grandchild(Grandchild, Grandparent) :-

grandparent(Grandparent, Grandchild).

% Uncle rule

uncle(Uncle, Person) :-

male(Uncle),

sibling(Uncle, Parent),

parent(Parent, Person).

% Aunt rule

aunt(Aunt, Person) :-

female(Aunt),

sibling(Aunt, Parent),

parent(Parent, Person).

Input / Queries

parent(john, peter).

Output:

```
| ?- consult('C:/Users/MOSHIYA/Downloads/exp 23.pl').  
compiling C:/Users/MOSHIYA/Downloads/exp 23.pl for byte code...  
C:/Users/MOSHIYA/Downloads/exp 23.pl compiled, 59 lines read - 3730 bytes written, 15 ms  
yes  
| ?- parent(john, peter).  
true ? c
```

Result

Thus, the family tree was successfully implemented using Prolog, and various family relationships such as parent, child, sibling, grandparent, uncle, and aunt were successfully determined using Prolog queries.